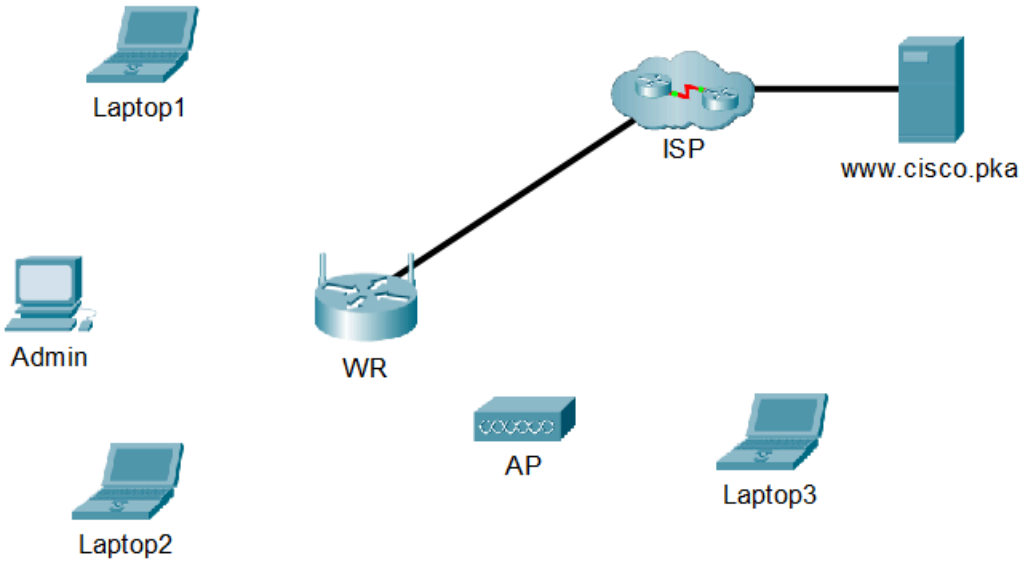
# Lab 5.1 - Configure a Basic Wireless Network (PT)

This activity comes with an accompanying Packet Tracer file with a partially configured network. Make sure to download the Packet Tracer file from the Animospace assignment page.




## Addressing Table

| Device | Interface | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|---|
| R1 | Internet | 209.165.200.225 | 255.255.255.252 | 209.165.200.226 |
| | LAN | 192.168.0.1 | 255.255.255.0 | N/A |
| www.cisco.pka | NIC | 209.165.201.2 | 255.255.255.224 | 209.165.201.30 |
| Admin-PC | NIC | DHCP Enabled | | |
| LaptopA | NIC | DHCP Enabled | | |
| LaptopB | NIC | DHCP Enabled | | |
| LaptopC | NIC | DHCP Enabled | | |

## Objectives

- Connect to and configure a wireless router
- Connect a LAN devices to the wireless router

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- Add an AP to the network to extend wireless coverage
- Update default router settings

## Introduction

In this activity, you will configure a wireless router and an access point to accept wireless clients and route IP packets. Furthermore, you will also update some of the default settings.

## Instructions

## Part 1: Connect to and Configure a Wireless Router

This lab uses a generic wireless router that is modeled after a Linksys WRT series wireless home router. In this section, you will go through the motions of connecting to a new wireless router using a wired host and accessing its web GUI for configuration.

### Step 1: Connect a management host to a wireless router.

Connect **Admin-PC** to any of the Ethernet ports of **WR** using a straight-through through the Ethernet ports.

**WR** will act as a switch to the devices connected to the LAN and as a router to the internet. **Admin** is now connected to the LAN (**GigabitEthernet 1**).

### Step 2: Configure Admin to use DHCP.

To reach the **WR** management page, **Admin** must communicate on the network. A wireless router usually includes a DHCP server, and the DHCP server is usually enabled by default on the LAN. **Admin** will receive IP address information from the DHCP server on **WR**.

a. Click **Admin**, and select the **Desktop** tab.

b. Click **IP Configuration** and select **DHCP**.

| | |
|---|---|
| What is the IP address of the computer? | 192.168.0.100 |
| Which is the subnet mask of the computer? | 255.255.255.0 |
| What is the default gateway of the computer? | 192.168.0.1 |

c. Close the **IP Configuration** window.

### Step 3: Connect to the WR Web Interface.

a. In the **Desktop** tab on **Admin-PC**, choose **Web Browser**.

b. Enter **192.168.0.1** in the URL field to open the web configuration page of the wireless router.

c. Use **admin** for both the username and password.

d. Under the Network Setup heading on the **Basic Setup** page, scroll down to the section showing the DHCP server settings and observe the indicated IP address range

| | |
|---|---|
| Does the indicated range match the address assigned to the Admin-PC? | Yes |

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### Step 4: Configure the Internet Port of WR.

When connecting to an ISP or any external network, the Internet port of the wireless router needs to configured with address settings assigned by the external network. In this step, **WR** is configured to route the packets from the wireless clients to internet. You will configure the **Internet** port on **WR** to connect to the internet.

a. Under the **Internet Setup** at the top of the **Basic Setup** page, change the Internet IP address method from **Automatic Configuration – DHCP** to **Static IP**.

b. Type the IP address to be assigned to the Internet interface as follows:

**Internet IP Address**: 209.165.200.225

**Subnet Mask**: 255.255.255.252

**Default Gateway**: 209.165.200.226

**DNS Server**: 209.165.201.1

c. Scroll down the page and click **Save Settings**.

d. To verify connectivity, open a new web browser and navigate to **www.cisco.pka** server.

**Note** It may take a few seconds for the network to converge. Click **Fast Forward Time** or **Alt+D** to speed up the process.

### Step 5: Configure the WR SSID.

In this step, you will only configure the wireless settings of **WR** for 2.4 GHz.

a. Navigate to the **WR** GUI interface at **192.168.0.1** in a web browser on **Admin**.

b. Navigate to **Wireless > Basic Wireless Settings**.

c. Change **Network Name (SSID)** to **aCompany** for only 2.4 GHz. Notice that SSIDs are case-sensitive.

d. Change the **Standard Channel** to **6 - 2.437GHz**.

e. For this activity, disable both 5 GHz frequencies. Leave the rest of the settings unchanged.

f. Scroll to the bottom of the window and click **Save Settings**.

### Step 6: Configure wireless security settings.

In this step, you configure the wireless security settings using WPA2 security mode with encryption and passphrase.

a. Navigate to **Wireless > Wireless Security**.

b. Under the 2.4 GHz heading, select **WPA2 Personal** for the Security Mode.

c. For the Encryption field, keep the default **AES** setting.

d. In the Passphrase field, enter **Cisco123!** as the passphrase.

e. Click **Save Settings**.

f. Verify that the settings in the **Basic Wireless Settings** and **Wireless Security** pages are correct and saved.

## Part 2: Connect the Wireless Clients.

Mobile clients may be connected to a wireless LAN by adding a wireless NIC and configuring wireless and security settings to match those configured for the network.

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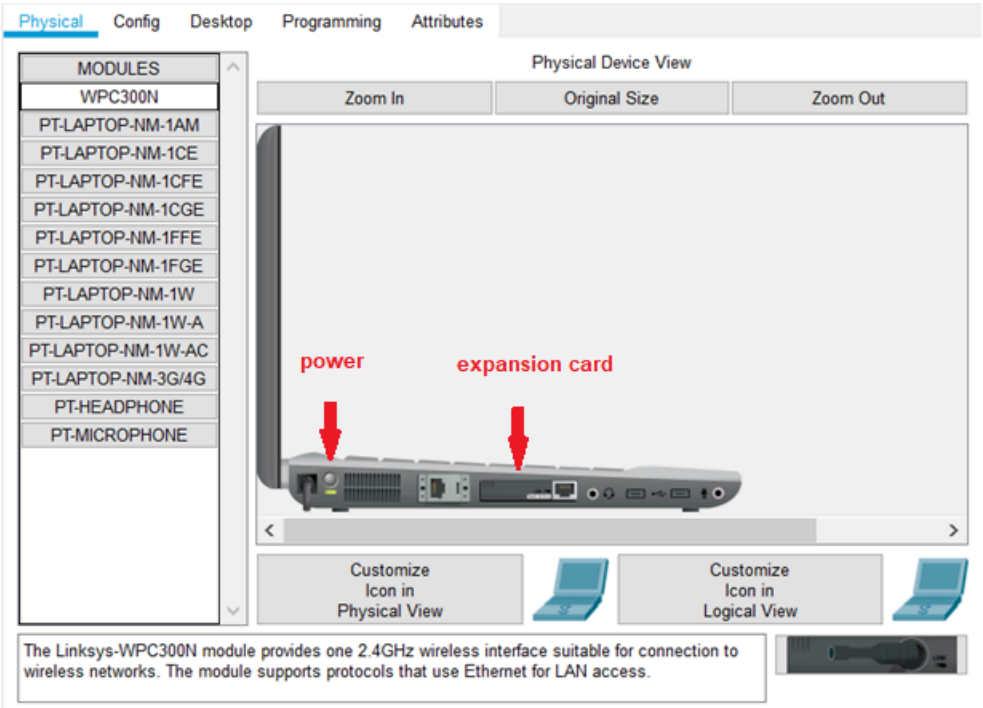
## Step 1: Prepare a Laptop for wireless connectivity

In Packet Tracer, a laptop does not have a wireless NIC by default; hence using one in a simulated WLAN topology requires a wireless NIC to be manually installed. In this Step, you will be installing a wireless NIC on Laptop1 as practice. (Laptops 2 and 3 have already been preinstalled with a wireless NIC)

a. Open **Laptop1** and select the **Physical** tab

b. Turn off the laptop by clicking on its power button in the device image. This is the round gray button with a yellow LED below located beside the laptop cooling vents. The yellow LED should turn off to indicate that the laptop is now powered down.




c. Remove the wired NIC expansion card by clicking and dragging it away from the laptop then dropping it into the modules pane on the left.

d. Select the WPC-300N module from the Modules pane. This is an 802.11n compatible wireless NIC. A short description and a picture of the module should appear below.

e. Install the wireless NIC module by clicking its picture beside the description. Drag and drop it into the expansion card slot of the laptop.

f. Turn on the laptop by clicking on the power button again.

## Step 2: Connect Wireless Clients

a. On **Laptop1**. Select **Desktop** tab then click **PC Wireless**.

b. Select the **Connect** tab. Click **Refresh** as necessary. A list of wireless network SSIDs should appear in the list.

c. Select the Wireless Network Name **aCompany**.

d. Enter the passphrase configured in the previous step. Enter **Cisco123!** In the pre-shared key field and click **Connect**. Close the PC Wireless window.

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e. Open a web browser and verify that you can navigate to **www.cisco.pka** server.

f. Repeat the above steps to connect **Laptop2** to the wireless network.

## Part 3: Connect Clients to an Access Point

An access point (AP) is a device that extends the wireless local area network. An access point is connected to a wired router using an Ethernet cable to project the signal to a desired location.

### Step 1: Configure the Access Point.

a. Connect **Port 0** of **AP** to an available Ethernet port of **WR** using a straight-through Ethernet cable.

b. Click **AP**. Select the **Config** tab.

c. Under the INTERFACE heading, select **Port 1**.

d. In the SSID field, enter **aCompany**.

e. Select **WPA2-PSK**. Enter the passphrase **Cisco123!** In the Pass Phrase field.

f. Keep **AES** as the default Encryption Type.

### Step 2: Connect the Wireless Clients.

a. Open **Laptop3**. Select **Desktop** tab. Click **PC Wireless**.

b. Select the **Connect** tab. Click **Refresh** as necessary. Select the Wireless Network Name **aCompany** with the stronger signal (Channel 1) and click **Connect**.

c. Open a web browser and verify that you can navigate to **www.cisco.pka** server.

## Part 4: Other Administrative Tasks

### Step 1: Change the WR Access Password.

a. On **Admin-PC**, navigate to WR GUI interface at **192.168.0.1**.

b. Navigate to **Administration > Management** and change the current **Router Password** to **cisco**.

c. Scroll to the bottom of the window and click **Save Settings**.

d. Use the username **admin** and the new password **cisco** when prompted to log in to the wireless router. Click **OK** to continue.

e. Click **Continue** and move on to the next step.

### Step 2: Change the DHCP address range in WR.

In this step, you will change the internal network address from 192.168.0.0/24 to 192.168.50.0/24. When the LAN network address changes, the IP addresses on the devices in the LAN and WLAN must be renewed to receive new IP addresses before the lease is timed out.

a. Navigate to **Setup > Basic Setup**.

b. Scroll down the page to **Network Setup**.

c. The IP address assigned to **Router IP** is 192.168.0.1. Change it to 192.168.50.1. Verify that DHCP pool IP address still starts at .100, and there are 50 available IP addresses in the pool.

d. Add **209.165.201.1** as the DNS server with the DHCP settings.

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e. Scroll to the bottom of the window and click **Save Settings**.

f. Note that the DHCP range of addresses has been automatically updated to reflect the interface IP address change. The Web Browser will display a **Request Timeout** after a short time.

Why do you think this happened?

| |
|---|
| This happened because our router IP changed and is no longer 192.168.0.1. The admin's IP address is not in the same subnet. Hence, it request timeout |

g. Close the **Admin-PC** web browser.

h. In **Admin-PC Desktop** tab, click **Command Prompt**.

i. Type **ipconfig /renew** to force **Admin** re-acquire its IP information via DHCP.

| | |
|---|---|
| What is the new IP address information for **Admin-PC?** | 192.168.50.104 |

j. Verify that you can still navigate to **www.cisco.pka** server.

k. Renew the IP address on other laptops to verify that you can still navigate to **www.cisco.pka** server.

l. Notice that **Laptop1** connected to the **AP** instead of **WR** this time.

What could be the reason for this?

| |
|---|
| This is because the AP offers a stronger signal strength compared to WR this time |

## Reflection

Would this method of using APs or wireless routers be a scalable solution to deploy a wide coverage WLAN to a large organization? Why or why not?

| |
|---|
| No, it is better to use enterprise-grade LAN controllers for large organizations. This method offers centralized configuration and management which makes it easier for the admin to add new access points or make changes to the network. Moreover, a large organization with numerous users should not share one password. An authentication server containing all the user credentials is needed. |

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